

Distance between a plane and point

Basic skills

Q1. a) $6x - 3y + z = -2$
 $\underline{n} = (6, -3, 1)$

b) $z = 5 + x - 3y$ $x - 3y - z = -5$
could have $\rightarrow \underline{n} = (1, -3, -1)$
 $\underline{n} = (-1, 3, 1)$

Q2. vector between points is $\left(\begin{smallmatrix} 5 \\ 4 \\ 2 \end{smallmatrix} \right) - \left(\begin{smallmatrix} -4 \\ -2 \\ 0 \end{smallmatrix} \right) = \left(\begin{smallmatrix} 9 \\ 6 \\ 2 \end{smallmatrix} \right)$ then $\left| \begin{smallmatrix} 9 \\ 6 \\ 2 \end{smallmatrix} \right| = \sqrt{9^2 + 6^2 + 2^2} = 11$ ← magnitude

Q3. $(\underline{r} - \underline{a}) \cdot \underline{n} = 0$ $\underline{a} = \left(\begin{smallmatrix} 2 \\ 5 \\ 0 \end{smallmatrix} \right)$ $\underline{n} = \left(\begin{smallmatrix} 2 \\ -5 \\ 1 \end{smallmatrix} \right)$

so $2x - 5y + z = d$ or $\underline{a} \cdot \underline{n} = d$

then sub $\underline{a}$ so $2 \times 2 - 5 \times 5 + 0 = d$

$d = -21$ so $2x - 5y + z = -21$

Q4. $\underline{r} = \left(\begin{smallmatrix} -1+t \\ 2-t \\ 2t \end{smallmatrix} \right)$ $x = t - 1$ $IN \ 5x - y + z = 33$
 $y = 2 - t$ $\Rightarrow 5t - 5 + t - 2 + 2t = 33$
 $z = 2t$ $\Rightarrow 8t = 40 \Rightarrow t = 5$

so $x = 4, y = -3, z = 10$ $(4, -3, 10)$

Q5. Plane: $x + y + z = 9$ Point: $(3, 4, 5)$

a) $\underline{n} = (1, 1, 1)$

so $\underline{r} = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3+\lambda \\ 4+\lambda \\ 5+\lambda \end{pmatrix}$

b) so $x = 3 + \lambda$, $y = 4 + \lambda$, $z = 5 + \lambda$

$\Rightarrow 3 + \lambda + 4 + \lambda + 5 + \lambda = 9 \Rightarrow 3\lambda = -3$

so intersects at $\lambda = -1$

for point $\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$

c) distance between $(3, 4, 5)$ and $(2, 3, 4)$
is magnitude of $(1, 1, 1)$ so $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

$\lambda \underline{n}$

d) $\underline{p} \cdot \underline{n} = 3 + 4 + 5 = 12$ $d = 9$ $|\underline{n}| = \sqrt{3}$

so formula $\frac{|12 - 9|}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \frac{3\sqrt{3}}{3} = \sqrt{3}$

Q6. $2x - y + 3z = 12$ $(0, 0, 0)$

line is

$$\vec{r} = \begin{pmatrix} 2\lambda \\ -\lambda \\ 3\lambda \end{pmatrix}$$

Formula:

$$\frac{|0 - 12|}{\sqrt{2^2 + 1^2 + 3^2}} = \frac{12}{\sqrt{14}} = \frac{6\sqrt{14}}{7}$$

so then for intersection

$$4\lambda + \lambda + 9\lambda = 12 \Rightarrow 14\lambda = 12 \Rightarrow \lambda = \frac{12}{14}$$

so on plane is $\frac{1}{14} \begin{pmatrix} 24 \\ -12 \\ 36 \end{pmatrix}$

and distance is magnitude of difference

$$\text{so } \frac{1}{14} \begin{pmatrix} 24 \\ -12 \\ 36 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} 24 \\ -12 \\ 36 \end{pmatrix}$$

$$\frac{1}{14} \left| \begin{pmatrix} 24 \\ -12 \\ 36 \end{pmatrix} \right| = \frac{12}{14} \left| \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \right| = \frac{12}{14} \sqrt{2^2 + 1^2 + 3^2} = \frac{12}{14} \sqrt{14} = \frac{6\sqrt{14}}{7}$$

Q7. Plane has $\underline{n} = (0, 0, 1)$ and so

Point: $(1, 2, 1)$ $z = d$ as goes $(s, t, 2)$

we have $z = 2$

Formula

$$\frac{|1 - 2|}{\sqrt{1^2}} = 1$$

For line $\underline{r} = \begin{pmatrix} 1 \\ 2 \\ 1+\lambda \end{pmatrix}$ so $1+\lambda = 2$
 $\lambda = 1$

so intersects at $\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$

so vector for point to intersection is

$$\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

magnitude of $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ so distance is 1

Advanced skills

Q8. $\underline{r} = \begin{pmatrix} t-1 \\ 2-t \\ 2t \end{pmatrix}$ plane $\underline{n} = \begin{pmatrix} -3 \\ 3 \\ k+2 \end{pmatrix}$

$$\text{so } -3x + 3y + (k+2)z = d$$

Point $\begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$ so $-9 - 3 + k + 2 = d$
 $d = k - 10$

$$\text{so plane } -3x + 3y + (k+2)z = k - 10$$

a) For no intersection no solution

to $-3t + 3 + 6 - 3t + (k+2)2t = k - 10$

$$2(k-1)t = -19 \text{ no solution if } k=1$$

b) If $k=1$ then $\underline{a} = \begin{pmatrix} -3 \\ 3 \\ 3 \end{pmatrix}$

$$\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 3 \\ 3 \end{pmatrix} = -3 - 3 + 6 = 0$$

So Parallel

c) $(5, -4, 12)$

$$\begin{pmatrix} 5 \\ -4 \\ 12 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

$$5 = -1 + t$$

$$-4 = 2 - t$$

$$12 = 2t$$

$t = 6$ solves all
so yes on line

d) $\underline{r} = \begin{pmatrix} 5 \\ -4 \\ 12 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 3 \\ 3 \end{pmatrix}$ intersect plane

$$-3x + 3y + 3z = -9$$

$$-3(5-3\lambda) + 3(-4+3\lambda) + 3(12+3\lambda) = -9$$

$$\Rightarrow 27\lambda = 18 \quad \lambda = \frac{2}{3} \text{ so at } \begin{pmatrix} 3 \\ -2 \\ 14 \end{pmatrix}$$

so between is $\begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}$ for distance $\sqrt{12} = 2\sqrt{3}$

e) using formula instead:

$$\frac{|-1 \times -3 + 2 \times 3 + 0 \times 3 - -9|}{\sqrt{3^2 + 3^2 + 3^2}} = \frac{18}{3\sqrt{3}} = \frac{18\sqrt{3}}{9} = 2\sqrt{3}$$

so yes